Solution Architecture

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer
HTML Form (indexnew.html) - User enters Life Expectancy, Expected Schooling, Mean Schooling, GNI Per Capita
Result Page (resultnew.html) - Displays HDI Score and Category

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Flask Application Layer
app.py - Handles URL routing, POST requests from user form
Routes: / (home page), /predict (prediction endpoint)

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No Auth Service **Open access web app** **No login required**	**Core Logic - ML Model** **Linear Regression model** **Predicts HDI Score (R2=97.73%)** **Classifies HDI Category** **Pickle model (HDI.pkl)**	**No External APIs** **All data from local Kaggle CSV dataset** **No payment/email service**

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Data / Storage Layer
CSV File: Human Development Index - Full.csv (195 countries)
Pickle File: HDI.pkl (trained Linear Regression model)
Label Encoder: label_encoder.pkl

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	*HTML/CSS templates for user input form and prediction result display. User enters 4 HDI indicators and views the predicted score.*	*HTML, CSS, Jinja2 (Flask templates)*
Flask Application Layer (API Gateway)	*Flask app.py handles all HTTP requests. Routes / renders home page, /predict processes POST request and returns prediction result.*	*Python 3.x, Flask Framework*
Core Logic Service (ML Model)	*Linear Regression model predicts HDI score from 4 input features. Model loaded from HDI.pkl file. R2 accuracy = 97.73%.*	*Scikit-learn, NumPy, Pandas, Pickle*
Data / Storage Layer	*CSV file stores HDI data for 195 countries. HDI.pkl stores the trained ML model. label_encoder.pkl stores the label encoder.*	*CSV File, Pickle (.pkl), Pandas*